

MODULE 1 · THE INTEGRATION FOUNDATION

Failure to Leverage + BPR as Remedy

Why a working backbone so often fails to pay — and the discipline that closes the gap

Friday, September 11, 2026 · 8:00–10:30 AM · 2 MTC Room 802

IE-GY 9113B — Systems Integration: From ERP to Agentic AI · Prof. Emanuele Cimica

Today: activate the note, hear a live case, then build

- 1 Activation 8:05**

Rebuild the week's spine from memory — the fast floor everyone joins from.
- 2 Reconnect + story 8:17**

Back to the backbone as single source of truth; Birmingham and Nike.
- 3 Guest — a live ERP case 8:27**

An industry partner on a real implementation he managed, then discussion.
- 4 Apply — 3 breakouts 8:57**

By team, you build the S4 piece: your context-and-diagnosis brief.
- 5 Prime Lab 1 9:59**

Diagnose & Map — control points, the friction chain, the data sandbox.
- 6 Homework + wrap 10:14**

Lab 1 pre-work · the S4 brief · team charter due end of today.

Rebuild the spine — out loud, from memory

Four questions. Answer before I do. This is the floor the whole working session stands on.

Hold the frame. One arc — integrate, automate, make intelligent, make agentic; Module 1 is the first word. Three threads — Customer, Operations & Supply Chain, Cash Management. Four archetypes for scale, one operating-model lens.

Q1 What are the two kinds of failure?

Q2 Why is ERP the hardest project a company runs?

Q3 What is the cure — and how literally do we take it?

Q4 How do we find where value leaks?

Q1 — Two failures, not one

Implementation failure

The system does not work.

It crashes, loses data, cannot go live, is abandoned. Everyone sees it; everyone agrees it failed. Comparatively rare.

Failure to leverage

The system works — and value never arrives.

It processes transactions, stays up, passes its tests, and the organization is no better off. Money spent, technology functions, no payoff. This is the normal outcome.

>70%

of recent ERP initiatives will fail to fully meet their original business-case goals by 2027 — up to a quarter catastrophically (Gartner).

Q2 — Why ERP is the hardest thing a company does

The largest, least-rehearsable project most firms ever run — and the software is rarely the root cause. The failure clusters in human and organizational choices.

Legacy, multi-system reality

A strong core backbone that must integrate with specialized systems — not absorb them. The seams, not the systems, are where value leaks.

The delivery ecosystem

Integrators, internal IT, the vendor, and the business stakeholders who own the process. Choosing a partner is high-leverage; lowest price costs more later.

Retained design authority

Lean too hard on the vendor — or accept the platform's defaults — and you abdicate choices only the business can make. Stay the author of how your company works.

Integration on-ramps

Forms, email, file transfer, XML, APIs. Natural on-ramps raise adoption; miss them and work routes around the backbone, re-creating silos.

Transformation, not IT project. The technology is the easy part; the organization is the hard part.

Q3 — Don't automate, obliterate — but selectively

Hammer, 1990: stop paving the cow paths — redesign the work from a blank sheet before you automate it. But **“obliterate” is not “redesign everything.”** Concentrate total redesign where value is created or destroyed; adopt the standard where it isn't.

And decide the hard things early. Suppliers, ecosystem, requirements — the costliest to reverse, so make them when they are cheapest.

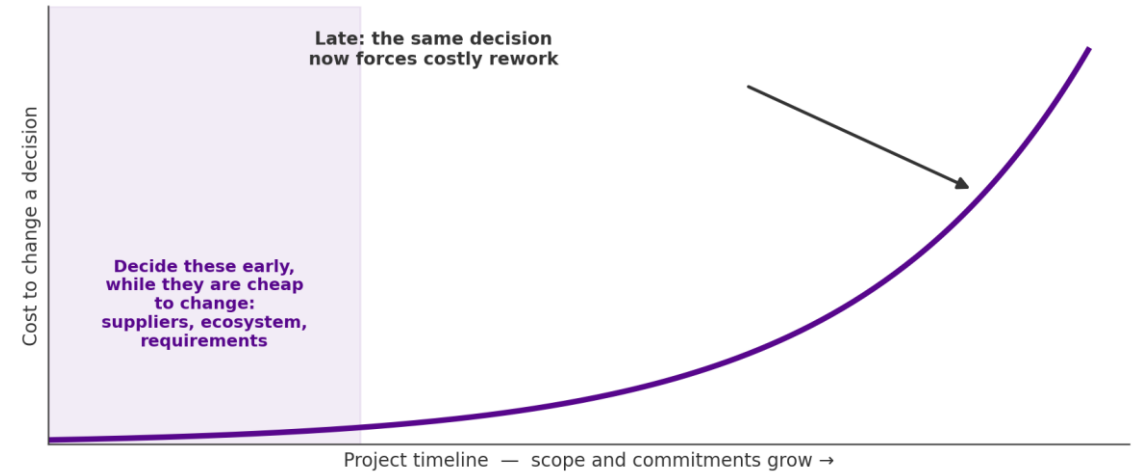


Figure 1 — the cost of changing a decision rises steeply as scope and commitments grow. Adapted from Boehm's cost-of-change curve (1981).

The deeper target: replace batch-push planning (overnight plans, morning firefighting) with pull-based, event-driven operations — decisions made at the point of variance. That is the redesign that makes automation, and later agents, actually pay.

Q4 — The Four Lenses: where value leaks

Four angles on any step, because each catches waste the others miss. Run all four; they overlap on purpose.

1

Frustration

The human signal — irritation marks broken handoffs and senseless rules. Rings Customer.

Customer

2

Time

Waiting, queuing, and above all rework. Rings Operations & Supply Chain.

Operations

3

Cost

Money consumed without value — duplication, manual reconciliation. Rings Cash.

Cash

4

Quality

Errors and defects that force the rework and the cost — one fix pays three times.

(all three)

Diagnose before you redesign; redesign before you automate. This is the engine of your Lab 1 pre-work — tag every leak you find by lens.

The spine, assembled

The order of operations this week builds — hold it for the breakouts:



The manager's job, reframed. Not “which tool speeds this up?” but the prior question — should this process exist in this shape at all?

Next, the whole toolkit from the note in one view — then two cases that fell into the trap.

Everything the note gave you — before we see it live

Not to re-learn — just to have it all on the board. A live case will exercise these next, and then you will put them to work on your own company.

Failure to leverage — it works, yet the value never arrives — the usual outcome, not the exception.

Legacy, multi-system reality — a strong core plus specialist systems; the seams between them are where value leaks.

Delivery ecosystem & on-ramps — integrators, vendor, IT and business deliver it; natural on-ramps (forms, email, API) lift adoption.

Requirements & retained authority — functional vs. technical needs — and staying the author of how you work, not the vendor's defaults.

Obliterate, then automate — redesign the work before you automate it; automating a broken process just paves the cow path.

Obliterate selectively — total redesign only where value is made or lost; adopt the standard everywhere else.

Batch-push → event-driven — the deeper redesign — decide at the point of variance, not the morning after.

Big-Bang vs. progressive — phasing buys discovery and bounds the blast radius; a Big-Bang makes every hidden problem simultaneous.

The Four Lenses — Frustration, Time, Cost, Quality — run all four to find where a process leaks value.

Adoption over change management — design the people-side in from the start; those inside the process help reshape it.

RECONNECT + STORY

From the backbone to two failures that broke it

Carry Week 1's backbone forward — then two failures a generation apart.

Back to Week 1: the backbone as single source of truth

Single source of truth

One authoritative record every function can read and trust — the non-negotiable the whole enterprise runs on.

Control points

Where data is first created or a commitment is made — where the record is born.

Decision points

Where work is routed — approved, rejected, or sent one way rather than another. Rejection and rework are the costly pattern.

Hold this. Every case today is a story about what happens when the single source of truth is destroyed.

The leverage gap

Week 1 closed on one unsettling idea. This week dissects it.

A backbone that works

Processes transactions. Stays up. Passes its tests. Technically, a success.



A backbone that pays

Turns activity into value the organization can actually keep. Far rarer — and the whole point.

The distance between them is the leverage gap. Now watch two organizations fall into it — Birmingham and Nike — then hear it from someone who lived it.

Birmingham — the cutover that erased the books

April 2022

Birmingham City Council — the largest local authority in Europe — went live on Oracle Cloud Fusion, replacing a long-standing SAP system, to run finance and HR.

The choice

Rather than re-engineering finance and HR to fit a modern platform, the council bent the platform to its legacy ways of working.

What broke

It posted transactions incorrectly and could not produce reliable accounts — the council lost its single source of truth. Costs escalated far beyond a modest business case.

5 September 2023 — a Section 114 notice. The local-government equivalent of declaring it cannot balance its budget. Lose the single source of truth and you lose the ability to account for yourself.

Birmingham — and here it gets hard

What actually caused the Section 114 is genuinely contested — and the contest is the lesson.

The council

Attributed the notice to its long-running equal-pay liability and an in-year budget gap. Its statement did not name Oracle at all.

Audit Reform Lab

Reading the same accounts, argues nearly the reverse — that “Oracle seems the more likely explanation of Birmingham's problems.”

The external auditor

A middle line — the Oracle failure as a contributory factor, faulting governance and optimism rather than the core technology.

More recently

Some experts argue the council may never have been truly bankrupt, on figures they call misstated; the commissioners reject that reading.

Why can experts still disagree? Because the failed system left the council unable to produce reliable numbers about its own position. Lose the single source of truth and you lose the ability to account for yourself.

Nike, 2000 — the same trap, in the clear

An i2 demand-planning engine bolted onto unredesigned processes, under-tested, rushed live in a Big Bang with under-trained planners.

What broke

Forecast errors flowed straight into factory orders. **Too many Air Garnetts; too few Air Jordans that were actually selling.** Orders ignored, duplicated, and deleted — planners could not reconstruct what they had told factories to make.

What it cost — the damage landed in the financials

~\$100M

in lost sales Nike publicly attributed to the episode

~20%

one-day drop in the stock, on a profit warning

~\$400M

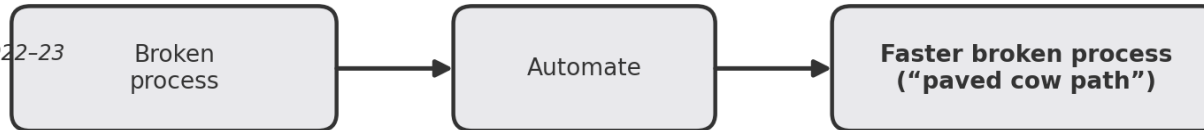
the broader supply-chain program it was measured against

The recovery proves the rule: Nike retrained planners and moved short-range planning back into its SAP backbone, grounded in actual orders — a move its CIO said let it “simplify some of our integration requirements.”

Two cases, one trap

The trap

Nike 2000 · Birmingham 2022-23



The cure

— Hammer 1990



Figure 2 — automate first and you pave the cow path; diagnose, obliterate, then automate, and you close the leverage gap. Adapted from Hammer (1990).

The rhyme is exact

Both treated a process-redesign problem as a software install.

Both automated before they obliterated.

Both went live in a Big Bang.

Both reached for one truth — a Unification ambition — without paying the standardization tax that earns it.

Lost sales for Nike; an unbalanceable budget for Birmingham. Two cases we read — now one we hear, from someone who ran it.

GUEST LECTURE · A LIVE ERP CASE

A real implementation — from someone who ran it

On your final-project executive review panel. He'll walk one real implementation he managed — the third case of the day, told live.

~20 minutes, then discussion. Listen for where value leaked, what he obliterated vs. paved, and how he carried people through the change.



Torrence Smith –
Global Enterprise
Architect &
Technology
Executive

Fantasy versus Reality

Fantasy

- The system will be composed of a single enterprise solution
- You'll be surrounded by skilled generalists
- The charter and major milestones will be clearly understood
- Your first job is to deliver a project

Reality

- You'll likely be responsible for rationalizing multiple systems in disconnected silos
- You'll be surrounded by specialists of varying experience and competencies
- You may be responsible for creating and socializing the charter and major milestones
- Your first job is to model the process

→ **CONNECTS TO WEEK 2** “Who's in control?” and “model the process first” are the week's retained-design-authority and business-first, data-last principles — the exact stance Lab 1 asks you to take. Disconnected silos = the legacy, multi-system reality where value leaks at the seams.

Mental model definition, from general to specific

- West Point Chemistry Class: It's about the framework. Resolve the formula before you add the numbers
- Big strategic idea = mental models
 - “First principles” thinking
 - Global enterprise view
 - They scale, they're elastic
 - Timeless: true 30 years ago, true today
 - Simplify complexity

→ **CONNECTS TO WEEK 2** Resolve the framework before the numbers mirrors the note's order: the managerial frame (why a working system fails to leverage) comes before any figure. “Applies regardless of scale” is the four archetypes — the same lenses read a CPG on SAP or a Series-B on Stripe.

Mental Model: Shea's Law

“The ability to improve a design occurs primarily at the interfaces. This is also the prime location for screwing it up.” —David Akin, Akin’s Laws of Spacecraft Design #15

- Interfaces can determine evolvability + interoperability
- Components saturate, interfaces compound
- Failures emerge at the boundaries
- Integration allows you to gain the resources of 3rd parties
- Keystone Actors + Pareto’s Law: Not every process is equally important. **Prioritize integration** - Application Programming Interface



→ **CONNECTS TO WEEK 2** “Failures emerge at the boundaries” is the note’s point that value leaks at the seams between systems — the integration on-ramps. Keystone Actors + Pareto is obliterate selectively: concentrate the redesign on the few processes where value is created or destroyed.

Use Case: Scrap Metal Recycling Industry

- Major global export business
- The arbitrage model
- Just in time inventory
- Provided shredded metal to foundries
- Non-ferrous = business within the business
- Profit measured in pennies



→ **CONNECTS TO WEEK 2** A real, sourced case — the kind the note treats as primary. Its stakes live on two of the three threads: Operations & Supply Chain (just-in-time material flow) and Cash Management (pennies-per-pound margins that only pay at velocity).

Non-ferrous Business

- Yards = a factory without a roof
- The non-ferrous business model: high mix, high volume, comparatively high margin
- Different materials, margins, fractions, and packaging
- Supply: joint venture, dealers, peddlers
- Packaging: loose, boxed, baled, palletized
- Profit measured in pennies, \$.20-30 / lb. of margin



Non-ferrous materials: copper, brass, aluminium, tungsten, lead.

New England: local symptoms of a global problem

96 days

Order-to-cash, average

51 days

Time to ship, average

51%

On-time delivery

- Global contracts held in local legacy systems
- North America and Singapore duplicate contracts
- No deposit visibility > trigger to begin the process
- Rework from rolled steamship bookings
- Lack of shipment staging
- 66% of our recycling export was to China*



→ **CONNECTS TO WEEK 2** This is the leverage gap made concrete: a business that runs but does not pay. O2C and time-to-ship are Time-lens leaks; OTD is a Quality leak; duplicate contracts and no deposit visibility are the lost single source of truth.

O2C — End to End Process Flow

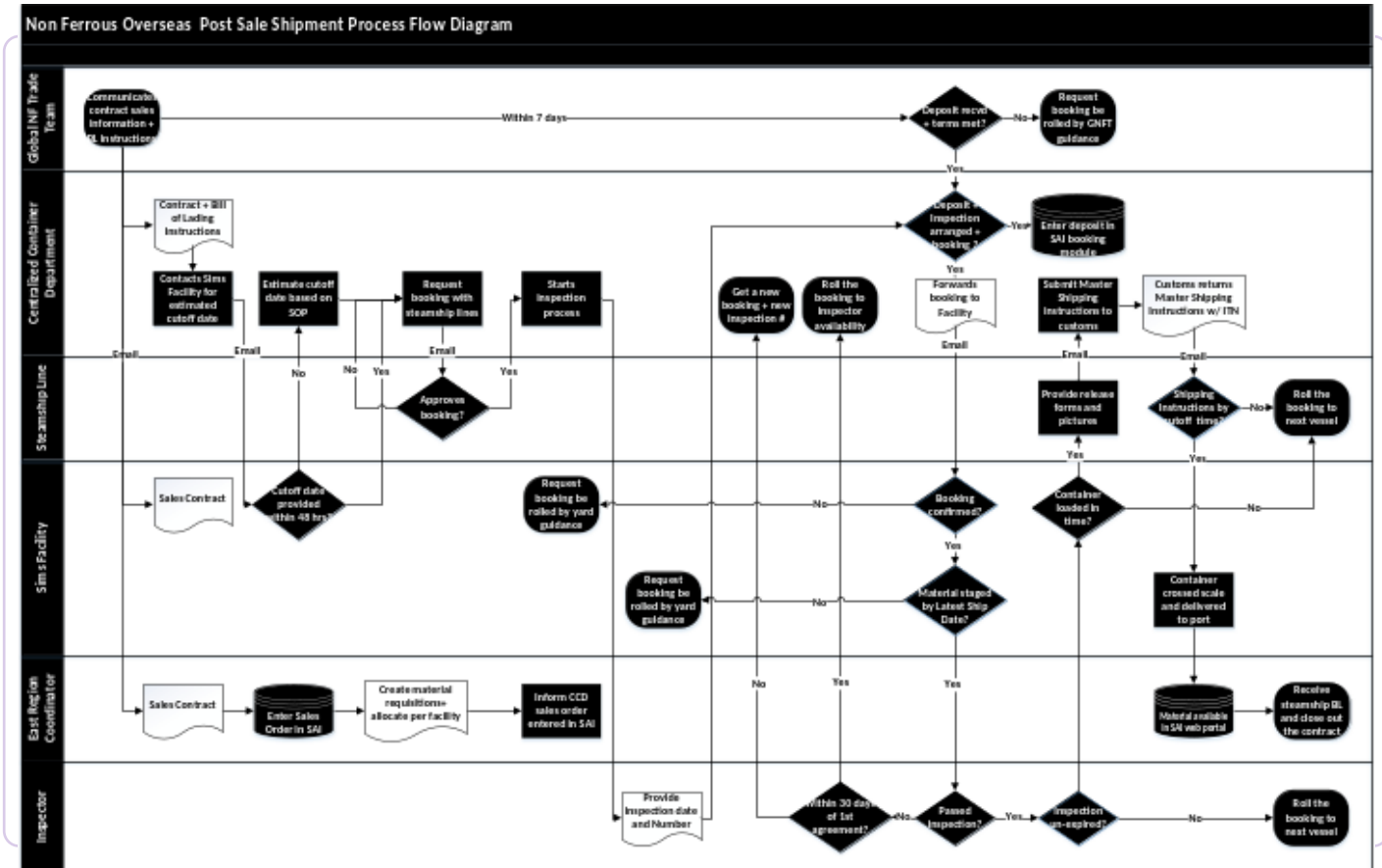
Time/ Cost: Where does the work stop flowing?

- 72 emails to process a contract
- A room filled with filing cabinets

Frustration/ Quality: Who is affected by the problem?

- Long hours
- Poor visibility + infighting

****No single source of truth** – rework process instead of a production process**



→ **CONNECTS TO LAB 1** This swimlane map is exactly the Lab 1 deliverable — a real end-to-end process with its control points, decision points, and rejection/rework paths. “Rework instead of production” is the friction chain the lab asks you to trace to a board-level consequence.

Stakeholder Analysis

Stakeholder	Key interests	Issues	Current Status	Interest	Power	Message	Action
Centralized Container Department	Efficiency	Complexity, lack of visibility	For	High	Medium	Reduced errors, easier work, reduced overtime	Personnel training, system trouble-shooting
NAM Trade	Inventory visibility, sales recognition	Poor visibility into sales, inventory	Neutral	Medium	Medium	Better visibility, improved inventory	MR discussion
Global Non-Ferrous Trade	Access to data	Lack of integration, manual processes	For	Medium	Medium	Simplified processes, improved service level	Beta testing
Finance	Order to cash, profitability	Excess time receivables, high # of claims	For	High	High	Reduced order to cash, reduced claims, reduced labor	Share SOW functionality
C Suite	Material velocity, market share	Difficulty to scale, high # of claims	For	High	High	Improved OTD, O2C, scalability	Delivery roll out plan
Consignees (customers)	Document availability, On time delivery	Late shipments, incomplete documents	For	Low	High	Improved OTD, O2C, document delivery, reduced errors	Implement the system
CTO	Compatibility to ERP	Need for replacement w ERP	Neutral	Low	High	Specialized function, full interoperability	Capability presentation
Operations	Ease of use	Complexity, lack of visibility	Neutral	Low	Medium	Easier work, improved visibility	Personnel training
Internal Audit	Compliance	Lack of access controls	Neutral	Low	High	BSM possesses adequate internal controls	Schedule IA review

→ **CONNECTS TO WEEK 2** This is the ERP delivery ecosystem from the note — every actor whose interest and power shape the build. Reading power against interest is how conscious partner selection and adoption over change management are decided.

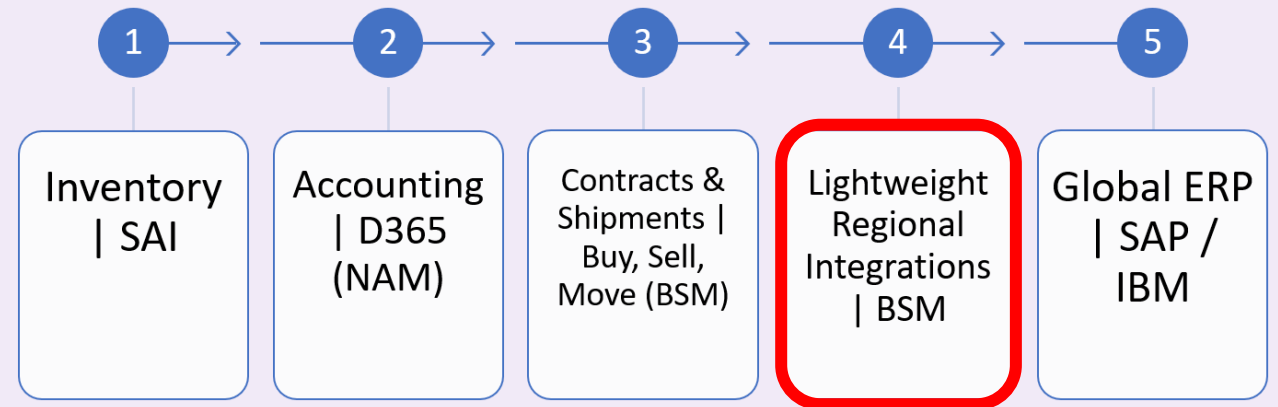
Control Plan

#	Sequence	Control Point	Risk	Type	Owner	Evidence
1	Order entry	CCD reviews orders for content	Missing contract information	Detect	GNFT	Contract number
3	Booking submitted	CCD requests cutoff date from yard; Accounting reconciliation	Blind booking; Rolled booking	Prevent	CCD	Booking number
4	Inspection scheduled	CCD stacks orders for inspector availability	Limited inspector availability; Failed inspection	Prevent	CCD	Inspection certificate; Inspection number
5	MRs allocated	CCD receives error in SAI	Contract allocated to yard with no inventory	Detect	Trade	MR Number
6	Deposit received	GNFT publishes deposit report; Control total in SAI	Delay in receipt of deposit; Delay in communicating deposit	Detect	GNFT	Fund transaction
7	Material staging	Sales release at scale	Inadequate material quantity	Detect	Yards	Sales release; Scale ticket
8	Master shipping instruction AES	File a dummy booking	Error in shipping documents; Not filing 24 hours before cutoff at terminal	Prevent	CCD	Master shipping instruction w/ ITN
9	Material In gate	Terminal verifies documents	Error in shipping documents; Ingating late	Detect	Yards	Terminal Interchange Receipt
10	Material loaded on steamship	Steamship loads container	Ingating late	Detect	CCD	Bill of lading number

→ **CONNECTS TO LAB 1** These are control points (★) — where the business commits or data is created — exactly what Lab 1 asks you to mark on your own map. Each risk/type/owner/evidence row is the governance that attaches at a control point.

Use Case: Tech Stack

- Region specific ERPs
- Finance and inventory in separate platforms
- Global Contracts Management from manual > automated
- SAP in global hubs
- SAP design initially without overall architecture
- Integration provided by something that wasn't on the original roadmap*



→ **CONNECTS TO WEEK 2** Tooling in this course is outcome-first: a Default no-code stack, an Enterprise AWS path, or a faculty-approved Custom stack — graded on business outcomes and solution design, never complexity. BSM vs SAP is that same build choice at enterprise scale.

BSM vs SAP (IBM Integrator)

BSM

- Software first logistics company
- Robust API w/ Lots of on-ramps (email, Jet reports, FTP, XML)
- Flexible construction > rework w/ automated audit trail
- Bottom up planning w/ extensive stakeholder inputs
- Incremental Go-live

SAP (IBM Integrator)

- Finance focuses software company
- Primary integration through an API + large integration team
- Rigid construction
- Top down planning w/ no previous process mapping > multiple replanning events
- “Big Bang” Go-live

→ **CONNECTS TO WEEK 2** This is the week's spine in one table: incremental vs. “Big Bang” go-live is the Nike/Birmingham lesson; “no previous process mapping” is paving the cow path; on-ramps + a robust API are the integration on-ramps; stakeholder inputs are adoption designed in from the start.

Results

4 months

Time to execute

96 → 58 days

Order to cash

51% → 85%

On-time delivery

34%

Monthly increase in
containers shipped

- Synchronization during China's "Operation National Sword"
- Quality/ Visibility: none → end to end
 - Exception + rework reporting
- Frustration/ human impact
 - Elimination of overtime
 - Facilitation of remote work during COVID

→ **CONNECTS TO WEEK 2** This is the leverage gap closed — the same before/after the note promises when redesign precedes automation and the rollout is incremental, not Big Bang. Note the win shows up on Cash (O2C) and Operations (OTD) together: one redesign, several threads paid.

GUEST CASE · CLOSE

Questions?

The context-and-diagnosis brief due this week asks you to do for your own company what this case does: name where the backbone fails to leverage, and the value at stake.

From his case to your diagnosis

~10 minutes. Put the live case through this week's tools — the same moves you're about to make on your own company.

1 Where did value leak?

Run the Four Lenses over his case — Frustration, Time, Cost, Quality. Which thread did it ring: Customer, Operations, or Cash?

2 Obliterate, or pave the cow path?

Did the redesign happen before the system went in, or did the old process get automated? Where would selective redesign have concentrated?

3 What made it stick — or not?

How were the people inside the process carried through the change? Where did adoption, requirements, or the rollout decide the outcome?

Carry this into your breakout. His case is a fourth data point beside Birmingham, Nike, and your own company — and a preview of what the review panel listens for.

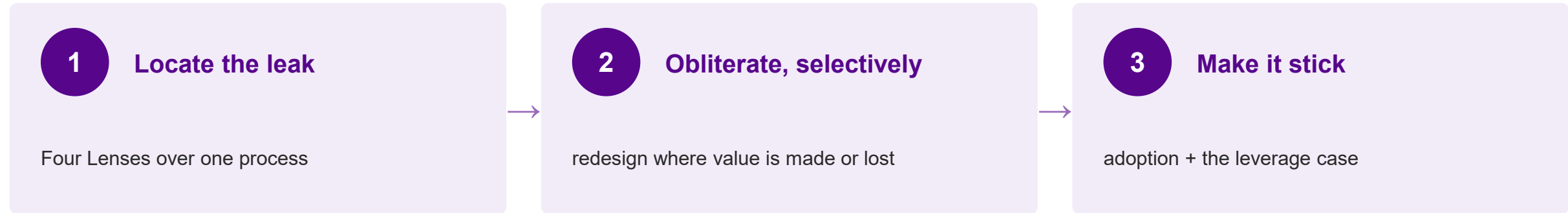
APPLY · BY PROJECT TEAM

Build the S4 piece of your board deck

A protected hour — three breakouts that assemble into your context-and-diagnosis brief. The note's toolkit is on the table; reach for it as you work.

Three breakouts, one deliverable

Everything you produce in the next hour assembles into your context-and-diagnosis brief — Section 2 of the board memo, slide S4 of your final deck.



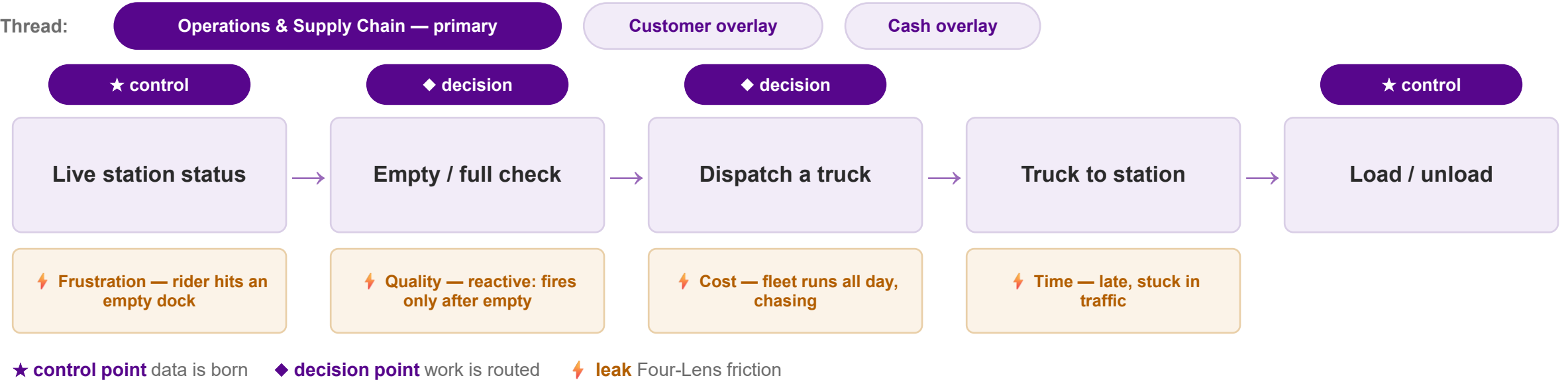
Work the steps — Each breakout gives you a short set of numbered steps — do them in order and you finish with the piece it names.

Entry ticket — Your study-question answers get your team into the room; the homework you did is the raw material.

By team — You build with your project team — the working session compounds team cohesion and capstone progress.

A backbone that sees every empty dock — and still can't prevent one

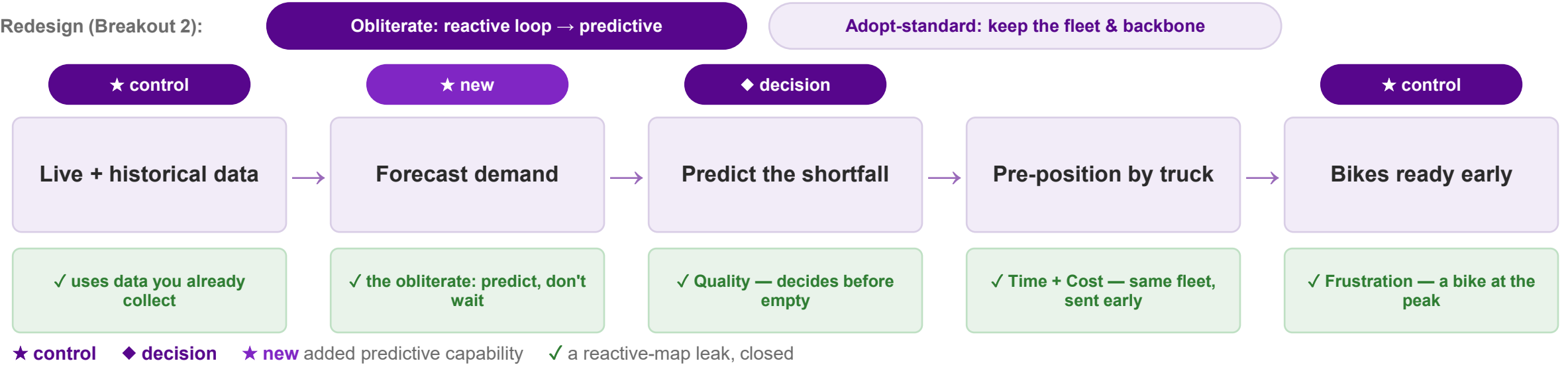
From Breakout 1 · Locate the leak: the ⚡ tags below are your Four-Lenses friction list. Citi Bike (operated by Lyft) rebalances bikes by truck; the dock-status backbone works, but the process is reactive — trucks dispatch only after a station empties.



The chain, to a number that matters: left unchanged, these leaks compound — reactive threshold → late truck → peak stockouts → lost rides + outsized rebalancing opex (illustrative: ~18% of peak trips hit a stocked-out station · ~\$4.50 per bike moved). One integrated service run as an unstandardised, reactive process — a working backbone that fails to leverage its own data.

Obliterate the reactive loop: predict demand, pre-position bikes

From Breakout 2 · Obliterate selectively (the redesign) and Breakout 3 · Make it stick (adoption, rollout, the case). Same fleet, same backbone — one decision redesigned; each ⚡ from the reactive map is now closed (✓).



Make it stick — Breakout 3, steps 1–2: ① who co-designs it — the drivers and dispatchers who live the Frustration lens; their traffic and hot-spot knowledge is the model's ground truth. ② the standardization tax — it strips their by-feel routing autonomy, so co-build the model with them, keep a manual override, and phase the rollout.

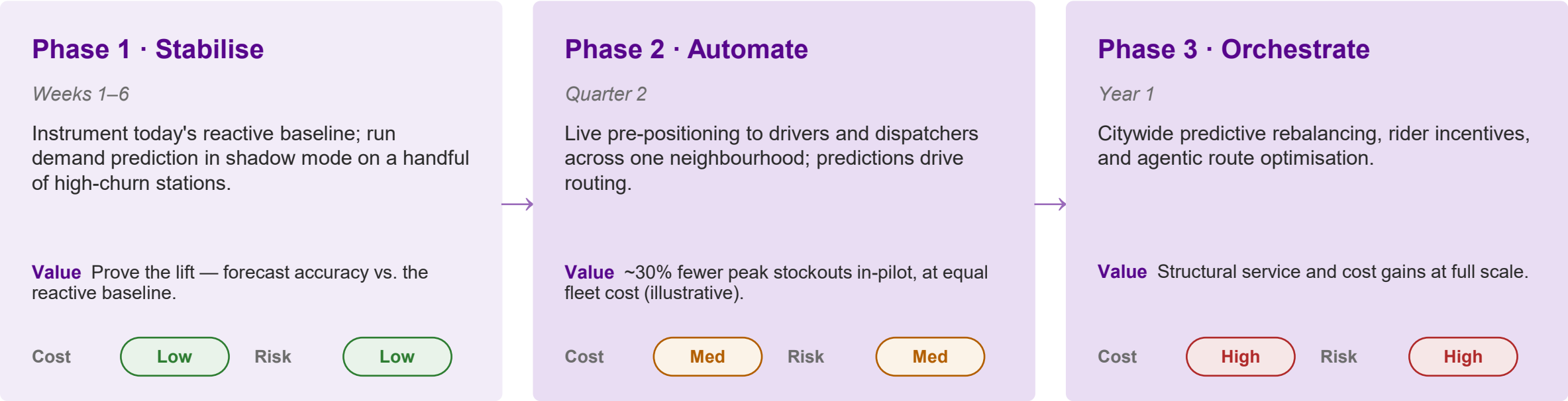
BREAKOUT 3 · THE LEVERAGE CASE — WHAT THE BOARD FUNDS

Fund a demand-prediction and pre-positioning capability — not more trucks — to cut peak stockouts and lower cost per bike moved: **an estimated 30–40% fewer peak stockouts at equal or lower fleet cost (illustrative).**

A leverage case, not an installation case — it redesigns the decision, not the plumbing. Figures illustrative.

Fund a small first phase — prove the lift, then scale

Beyond the three breakouts — the capstone's later roadmap (Milestone 3). *Three phases, decision-gated: spend least where certainty is lowest, and scale only once the lift is proven.*



THE ASK

Fund Phase 1 now. It is small, low-risk, and decision-gated: we prove the lift against the reactive baseline before committing a dollar to Phase 2. The board funds a capability that pays, in stages it can stop.

Value, cost and risk illustrative — quantified from the lab's synthetic sandbox.

Locate the leak

Take one end-to-end process on your company's backbone — Order-to-Cash or Procure-to-Pay — and run the Four Lenses over it.

WORK THROUGH IT, IN ORDER

- 1 Sketch the process end to end — a few boxes, from the first trigger to “done,” is enough.
- 2 Walk it with each lens in turn — Frustration, Time, Cost, Quality — and note at least one value leak under each.
- 3 Tag every leak with the thread it rings: Customer, Operations, or Cash.
- 4 Circle the overlap — the leak that shows up under two or three lenses. That is where one fix pays three times.

Output A Four-Lenses friction list for one process.

Time
12 min
+ 2 debrief

Debrief: two or three teams name their sharpest leak and the thread it rings.

What a good friction list shows

- 1 Every lens fired** If a lens found nothing, you looked at the database, not the work — go back to the people in the process.
- 2 Threads named** Each leak rings a thread: Frustration → Customer, Time → Operations, Cost → Cash. Quality rings all three.
- 3 The overlap found** The leak that shows up under Quality and again under Time and Cost is where redesign pays three times — that is your candidate.

Carry forward. Your sharpest, thread-crossing leak is the one Breakout 2 will decide whether to obliterate.

Obliterate — selectively

From your friction list, decide where to redesign — and, just as important, where to leave things alone.

WORK THROUGH IT, IN ORDER

- 1 Pick the one or two leaks where value is genuinely created or destroyed — and say why those, not everything.
- 2 For each, ask Hammer's prior question: should this step exist in this shape at all? Decide — total redesign, or incremental improvement.
- 3 Sketch the redesigned flow; name what you would obliterate vs. what you would adopt as the platform standard and leave alone.
- 4 Write one sentence: “Our backbone fails to leverage at ____, costing ____.”

Output A selective-redesign call: obliterate here, adopt-standard there.

Time

18 min
+ present

Present: each team shows its one redesign call — this is a capstone piece, so it is presented, not just discussed.

Selective, not total

- 1 Concentrated** The strongest calls redesign one or two things and leave working areas alone — obliterating everything fails for the opposite reason to Nike and Birmingham.
- 2 Standard where it doesn't drive outcome** Adopt the platform's standard for the parts that don't create or destroy value; spend redesign only where it does.
- 3 The rhythm, not just the steps** The deepest redesigns replace batch-push habits with event-driven decisions at the point of variance.

Retained design authority. You stayed the author of how the company works — you did not let the platform's defaults decide for you.

Make it stick — and make the case

Turn the redesign into something an organization will adopt and a board will fund.

WORK THROUGH IT, IN ORDER

- 1 Name who inside the process must help diagnose and reshape it — start from the Frustration lens. Adoption is designed in, not bolted on.
- 2 Name the standardization tax your redesign imposes — whose autonomy or roles it strips — and how you would soften it.
- 3 Decide how you would phase the rollout: progressive vs. Big Bang, and why.
- 4 Write the one-line leverage case — the sentence a board would fund. Not “install X” but “redesign Y so the business changes.”

Output An adoption + rollout note, and the one-line leverage case — not an installation case.

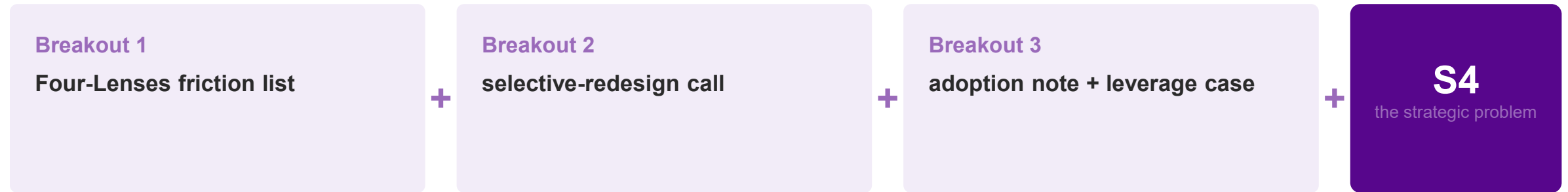
Time

15 min
+ present

Present: each team reads its leverage case aloud — one sentence a board would fund.

Your three outputs become the S4 brief

A board does not fund “buy this software.” It funds a redesigned process the software enables — a leverage case, not an installation case.



One page, due next week. A context-and-diagnosis brief on your backbone and where it fails to leverage — Section 2 of the board memo, and slide S4 of the final deck.

Section 2, with the operating-model read you began in Week 1, is the ground every later recommendation rests on.

NEXT WEEK

Lab 1 — Diagnose & Map a Process

You stop reading other people's failures and start mapping your own — spans Weeks 2–3.

Next week you stop reading failures and map your own

Lab 1 spans Weeks 2–3 — a consulting half before class, a build half in class. The order is deliberate: business first, data last.

1 Research the pain

Before class: pick your company and find its real friction and value leaks, through the Four Lenses.

2 Map the process

In class: map one end-to-end flow with its control and decision points.

3 Absorb the curveball

In class: a mid-build change — because real diagnosis survives a surprise.

4 Generate data last

In class: a synthetic-data sandbox — business logic first, data fitted to it.

That's the shape — the method is taught next week. The mapping vocabulary, the friction chain, and the sandbox are covered in full in the Week 3 note. Today, one thing to act on: your pre-work.

Your Lab 1 pre-work — a first pass, not the finished analysis

The consulting half: form a point of view before class. Rough is fine — you sharpen and map it in the lab.

1

Pick + a reality brief

Your Week-1 company, in one paragraph: archetype, integration stack (note any legacy), business model in a line, and the thread you suspect hurts most.

2

Research the pain

From public sources, find its real friction and value leaks — and tag each by lens (Frustration, Time, Cost, Quality) and thread.

3

Draft your hypotheses

Your top 2–3 suspected pain points, the flow you'd map, a first read on functional vs. technical requirements, and a rough first-guess chain.

It's a first pass — you refine it in class. In the lab you test it against a pattern library, map the process properly, and revise. So it never duplicates the lab: this is the raw material, the lab is the workshop. ~1.5–2 hrs, due before Week 3.

Where Lab 1 sits — and questions to argue

Lab 1 → your capstone, and the next labs

Feeds Section 3 — Backbone Assessment, and slide S5 of the final deck. It opens Milestone 1: problem diagnosis.

It also seeds the next labs: a stable, rules-based step becomes your Lab 2 automation; a step that needs judgment becomes your Lab 3 agent.

Bring rough answers — meant to be argued, not recited

1. What managerial failures most commonly precede a technical integration failure?
2. Define ‘failure to leverage’ in contrast to mere implementation failure.
3. Why must BPR precede automation — and what human risks appear when it is skipped?

Read which thread a failure lands on, and which lens surfaces it — that is the move your capstone asks you to make for your own company.

Homework — before we meet for Lab 1

1

Lab 1 pre-work (first pass)

Pick your company, research the pain via the Four Lenses, and draft your first-pass hypotheses (reality brief + suspected leaks). Due before the Week 3 lab.

2

Context-and-diagnosis brief

One page: your backbone and where it fails to leverage. This is Section 2 of the board memo and slide S4 of your deck.

3

Team charter

Due end of today — the one-page working agreement (choose open source template).

Set up your stack before Week 3. Lab 1 is a build week — get the tools ready for the stack you choose (Default: Miro/Lucidchart, Airtable/Sheets, Gemini; the AWS path; or your own) and work through their platform tutorials on your own before class.

THE SHIFT IN MINDSET

If you take one thing from this week, let it be the order of operations

1

Understand the terrain

2

Diagnose with the Four Lenses

3

Obliterate — selectively

4

Roll out progressively

5

Design adoption in from the start

A board funds a redesigned process, not a software install. Bring a leverage case, not an installation case.

Next week you map your own company's process — and the first thing you look for is the cow path.